Towards an Interpretable Urban Heat Island Foundation Model for Forecasting, Attribution, and Climate Adaptation Decision Support

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Extended Abstract

Urban Heat Islands (UHIs) intensify heat exposure in cities [1,4], yet current modelling approaches remain fragmented. Satellite-derived land surface temperature UHI (LST-UHI) and near-surface air-temperature UHI (AT-UHI) are usually studied separately because they reflect different surface and atmospheric heat mechanisms, although they provide complementary signals [2,3]. Existing machine learning models are often city-specific and mainly support static mapping or limited forecasting, which restricts transfer across climates and urban forms.

This PhD project investigates whether a unified, physically informed, and interpretable spatio-temporal foundation model can support transferable UHI forecasting and attribution. The proposed model is pre-trained on heterogeneous multi-city UHI datasets and adapted to forecasting, interpolation, attribution, and scenario simulation. The research focuses on how LST-UHI and AT-UHI can be jointly represented, which physical inductive biases improve cross-city generalization, and how uncertainty-aware outputs can inform adaptation.

The proposed UHIFM framework adopts a causal spatio-temporal Transformer encoder-decoder [5] with task-conditioned heads for LST-UHI and AT-UHI. Static urban features are grouped into morphology, activity, cooling, ventilation, and topography, and incorporated through feature-wise modulation [6]. A reliability-aware mechanism downweights incomplete feature groups, while a Tobler-inspired spatial module represents local dependence [7]. Training combines multi-horizon forecasting with masked temporal and spatial reconstruction. The overall research roadmap, spanning foundation model development, global benchmarking, and real-time decision support, is illustrated in Fig. 1.

Current work focuses on the first phase. Data collection and preprocessing have been completed for German air-temperature observations [8] and multi-city LST-UHI data, and the first PyTorch implementation is under development. Building on prior work on interpretable machine learning for UHI exposure and adaptive mitigation [9], the project aims to produce a transferable UHI

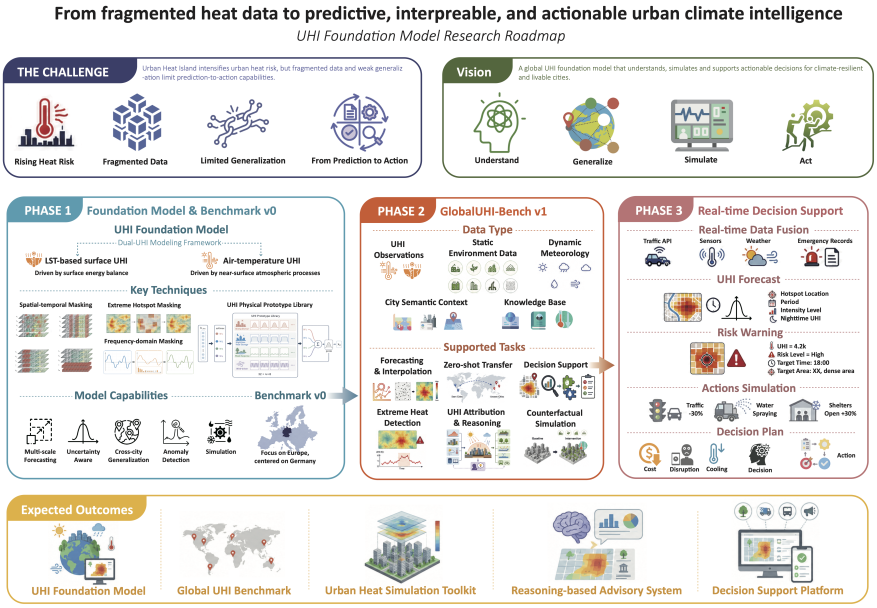


Fig. 1. UHIFM research roadmap across three phases: (1) foundation model and European benchmark, (2) GlobalUHI-Bench with cross-city transfer and attribution, and (3) real-time decision support.

benchmark and modelling framework linking forecasting accuracy with physical attribution and climate adaptation support.

Keywords: Urban Heat Island · Foundation Model · Spatio-temporal Forecasting · Interpretable Machine Learning · Urban Climate · GeoAI

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